

1 Proof of Cauchy-Binet

Last time we stated, but did not prove, the Cauchy-Binet formula. Here we give a proof which connects to the polynomial method. This way we also have a fully self-contained proof of the matrix tree theorem before we continue to use it today.

Theorem 1.1 (Cauchy-Binet). *Let $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times n}$ for $m \geq n$. Then:*

$$\det(AB) = \sum_{S \in \binom{[m]}{n}} \det(A_S) \det(B_S)$$

where A_S for $S \in \binom{[m]}{n}$ is the $n \times n$ submatrix consisting of the n columns indexed by S and similarly B_S is the n rows indexed by S .

We will first show two facts:

Fact 1.2. *Let $C \in \mathbb{R}^{m \times m}$. Then:*

$$\det(zI_m + C) = \sum_{S \subseteq [m]} z^{m-|S|} \det(C_{S,S})$$

Proof. Recall $\det(A)$ for any matrix A is a sum over permutations, with sign equal to 1 if the permutation is even (has an even number of inversions) or -1 otherwise. Further expand the permutations depending on whether they pick z or C_{ii} in each diagonal, and let S be the indices in which a permutation picks z . We can then group the permutations by their choice S . These picked diagonal entries do not affect the parity of the permutation since they contribute 0 to the number of inversions. The set of permutations picking a particular set of diagonals (and z in each one) therefore contribute exactly $\det(C_{\bar{S},\bar{S}})$. Taking complements gives the proof. $\square$

Fact 1.3. *Let $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times n}$ for $m \geq n$. Then:*

$$\det(zI_m + BA) = z^{m-n} \det(zI_n + AB)$$

Proof.

$$\det \begin{pmatrix} zI_m & B \\ -A & I_n \end{pmatrix} = \det \begin{pmatrix} zI_m + BA & 0 \\ -A & I_n \end{pmatrix} = \det(zI_m + BA)$$

where in the first equality we subtract B left multiplied on the second block of rows from the first block of rows, and this preserves the determinant. However, we can similarly do:

$$\det \begin{pmatrix} zI_m & B \\ -A & I_n \end{pmatrix} = \det \begin{pmatrix} zI_m & B \\ 0 & z^{-1}AB + I_n \end{pmatrix} = z^m \det(I_n + z^{-1}AB) = z^{m-n} \det(zI_n + AB)$$

where here we added $z^{-1}A$ left multiplied to the first row to the second row. $\square$

Putting these together, we get a proof of Cauchy-Binet.

Proof. By the second fact,

$$z^{m-n} \det(zI_n + AB) = \det(zI_m + BA)$$

Therefore the coefficient of z^{m-n} is equal. On the LHS, the coefficient is the constant term of the polynomial $\det(zI_n + AB)$ which is $\det(AB)$: just set $z = 0$. On the RHS, it is:

$$\sum_{S \subseteq [m]: |S|=n} \det((BA)_{S,S})$$

by the first fact. But $\det((BA)_{S,S})$ is $\det(B_S A_S) = \det(B_S) \det(A_S)$. □

2 Random Spanning Trees

Using Cauchy-Binet, the matrix tree theorem, and the mother-of-all lemma, we showed that λ -uniform spanning tree distributions are strongly Rayleigh, or equivalently that the generating polynomial of such distributions are real stable.

Today, we want to investigate properties of SR distributions. We will motivate our investigation with uniformly random spanning trees, and generate a Cayley-like formula that holds for general graphs.

2.1 Negative Correlation and the Negative Cylinder Property

A random variable $X \in \{0, 1\}^n$ is called pairwise negatively correlated if for all $i \neq j$:

$$\mathbb{P}[X_i = 1 \mid X_j = 1] \leq \mathbb{P}[X_i = 1]$$

or equivalently,

$$\mathbb{P}[X_i = 1, X_j = 1] \leq \mathbb{P}[X_i = 1] \mathbb{P}[X_j = 1]$$

in other words, conditioning on one element to be in the output only makes other elements less likely. It turns out this characterizes real stable multiaffine polynomials.

Lemma 2.1. *A multiaffine polynomial $p \in \mathbb{R}[z_1, \dots, z_n]$ is real stable if and only if for all $i, j \in [n]$ and all $x \in \mathbb{R}^n$ we have:*

$$p(x) \partial_{z_i} \partial_{z_j} p(x) \leq \partial_{z_i} p(x) \partial_{z_j} p(x)$$

Recall a multiaffine polynomial is one in which every variable has degree at most 1. Distributions over subsets always have multiaffine generating polynomials. As a corollary, we know that every SR distribution over subsets is pairwise negatively correlated. This is because $\mathbb{P}[X_i = 1] = \partial_{z_i} p(1)$, and $\mathbb{P}[X_i = 1, X_j = 1] = \partial_{z_i} \partial_{z_j} p(1)$, and $p(1) = 1$. In fact, it implies something stronger.

Definition 2.2 (Negative Cylinder Property). *A random variable $X \in \{0, 1\}^n$ has the negative cylinder property if for all $S \subseteq [n]$, we have:*

$$\mathbb{P}[X_i = 1 \forall i \in S] \leq \prod_{i \in S} \mathbb{P}[X_i = 1]$$

Lemma 2.3. *Every SR distribution over subsets has the negative cylinder property.*

We leave this as an exercise. Before proving this, note we can use this to give a bound on the number of spanning trees in the complete graph.

Lemma 2.4. *The complete graph has at least $(\frac{n}{2})^{n-1}$ spanning trees.*

Proof. Consider the uniform distribution over spanning trees on the complete graph, which is SR. By symmetry, every edge is included with probability $\frac{n-1}{\binom{n}{2}} = \frac{2}{n}$. By the NCP, each spanning tree is included with probability at most $\prod_{e \in T} \frac{2}{n} = (\frac{2}{n})^{n-1}$, which gives the claim. $\square$

We can compare this to Cayley's formula, which says there are n^{n-2} trees of the complete graph. Cayley's formula can also be proved using the matrix tree theorem.

Theorem 2.5. *The complete graph has n^{n-2} spanning trees.*

Proof. By the matrix tree theorem, $g_{\mathcal{T}}(1) = \frac{1}{n} \det(11^T/n + \sum L_e)$, which is the number of spanning trees. The Laplacian of the complete graph is just $nI - J$, where $J = 11^T$ is the all-ones matrix. The all-ones vector has eigenvalue 0, and every vector orthogonal to it has eigenvalue n . Adding $11^T/n$ changes the first eigenvalue to 1 and leaves the other $n - 1$ eigenvalues equal to n . Taking the product of the eigenvalues gives:

$$g_{\mathcal{T}}(1) = \frac{1}{n} \left(1 \cdot n^{n-1}\right) = n^{n-2} \quad \square$$

And of course there is a more general version of this statement which can be used to lower bound the number of spanning trees in any graph, or the number of objects in the support of any SR distribution:

Lemma 2.6. *Let $p_1, \dots, p_n$ be the marginals of a subset SR distribution $\mu : \{0, 1\}^n \rightarrow [0, 1]$. Then:*

$$\log(|\text{supp}(\mu)|) \geq \sum_{i=1}^n -p_i \log(p_i)$$

Proof. Let $X \sim \mu$. Then:

$$\begin{aligned} H(X) &:= \sum_{S \in \text{supp}(\mu)} \mu(S) \log \frac{1}{\mu(S)} \\ &\geq \sum_{S \in \text{supp}(\mu)} \mu(S) \sum_{i \in S} \log \frac{1}{p_i} && \text{By NCP} \\ &= \sum_{i=1}^n -p_i \log p_i. \end{aligned}$$

But entropy is also at most the log of the support size, giving the lemma. $\square$

2.2 Proof of Lemma 2.1

Let's talk a bit about the proof of this theorem of Brändén. We will only show that if p is real stable, then:

$$p(x)\partial_{z_i}\partial_{z_j}p(x) \leq \partial_{z_i}p(x)\partial_{z_j}p(x).$$

We leave the other direction as an exercise.

Proof. We know that the bivariate $q(z_i, z_j) = p(x_1, x_2, \dots, x_i + z_i, x_{i+1}, \dots, x_j + z_j, \dots, x_n)$ is real stable by closure under affine transformations. Since p is multiaffine, this polynomial is equal to

$$p(x) + \partial_i p(x)z_i + \partial_j p(x)z_j + \partial_i \partial_j p(x)z_1 z_2$$

So this is some bivariate real stable polynomial $a + bz_1 + cz_2 + dz_1 z_2$. You will prove on the homework that for such polynomials, we have $bc \geq ad$, which gives the claim. $\square$